Determine where $V(z)=z^4(2z-8)^3$ is increasing and decreasing

$$V'(z)=4z^3(2z-8)^3+6(2z-8)^2z^4$$

$$V'(z)=8z^3(z-4)^2(7z-16)$$

$$z=0, z=4, z=\frac{16}{7}$$

increasing → $(-\infty,0)\cup(\frac{16}{4},4)\cup(4,\infty)$

decreasing → $(0,\frac{16}{4})$